

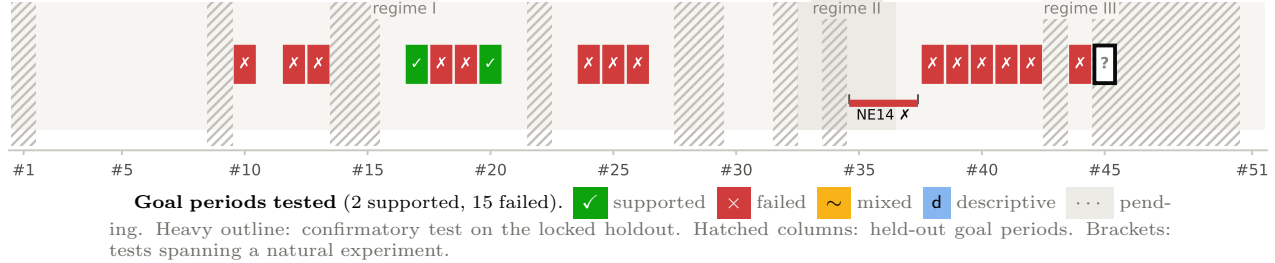
H02: Inferred couplings reflect real influence

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 0.5/5 Usefulness 1.5/5

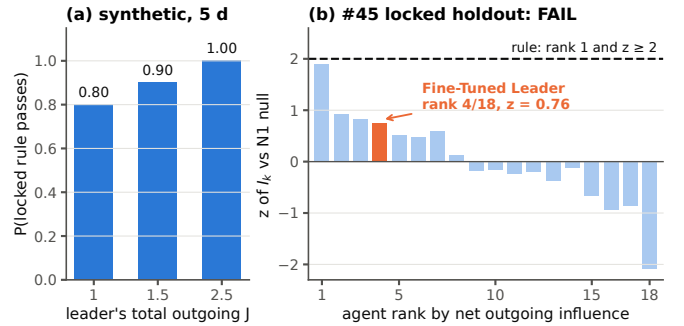
Bottom line. Inverse-Ising couplings from minute-level activity do not measure influence in an agent swarm: on corrected data they stay at the null floor, miss installed leaders and debate teams, and the apparent collective coupling is the operator's daily schedule.

level: hypothesis · pipeline checks 99% · updated 2026-10-04



WHAT HAPPENED

Directed kinetic-Ising couplings were fitted to 1-min activity spins in 21 non-holdout 5-day chunks (#10–#44), with block fields and surrogate nulls. Synthetic villages show the pipeline works (edge AUC 0.95; a leader with outgoing $J \geq 1.5$ found 90–100% of the time). On real data, pairwise couplings sit at the null floor in both goal modes, and #26's elected leader is not found. **Locked holdout #45 failed** (Fine-Tuned Leader 4/18, $z = 0.76$). **Round 1b** (corrected minute grid, which had dropped half of all events; day-edge-trimmed nulls): pairwise couplings are still null. The regime-III collective coupling of round 1 was the operator's day edges (significant in 2/8 regime-III chunks against 7/13 in regime I). #12's debate structure and #44's installed leader are invisible too. Post hoc, talk-spin couplings transfer across days in regime-III shared-objective weeks.



Observables. (a) Pass rate of the locked rule in synthetic villages sampled like #45 (18 agents, 5 days), by the planted leader's total outgoing coupling: a leader of strength 1.5 or more would almost surely have been found. (b) The real #45 holdout: z of net outgoing influence for every agent. The installed Fine-Tuned Leader ranks 4th at $z = 0.76$, and no agent reaches $z = 2$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A useful negative, now on clean data. Fitting an Ising model to who is active when, and reading off who leads, is the first thing a physicist tries on agent logs, and it does not work. Per-pair couplings from activity timing are noise, they missed a leader the operators installed (twice: #45 and #44), and they do not see a debate's team structure. Even the collective co-activation that looked like coupling is mostly the operator starting and stopping everyone each day. Anyone looking for which agent steers a swarm should look at messages: the only pairwise signal that survives is in who talks after whom, which fits H50's read-out-gated talk coupling. Mostly this tells you what not to compute.

PHYSICAL MODEL

Model 02 (kinetic Ising, primary); model 01 (equilibrium Ising) is the rival. Spin $s_i(t) = \pm 1$: agent i active (acts or talks) or not in 1-min bin t . h_i : bias; $\delta_i(b)$: field per (day, 30-min block), absorbing schedule and kickoffs; J_{ij} : lagged influence of j on i ; σ : logistic; I_k : net outgoing influence. Mean field: uniform coupling βJ_0 ; m : mean spin; q : mean spin variance; VR: variance of $\sum_i s_i$ over its independent value.

$$P(s_i(t+1) = +1 \mid \mathbf{s}(t)) = \sigma(2H_i(t))$$

$$H_i(t) = h_i + \delta_i(b) + \sum_j J_{ij} s_j(t)$$

$$I_k = \sum_{j \neq k} (J_{jk} - J_{kj})$$

$$N \text{Var}(m) = q/(1 - \beta J_0 q) \Rightarrow \beta J_0 = (1 - 1/\text{VR})/q$$

H02-couplings-are-real · generated 2026-10-04 10:55 by infra/summaries/build-summaries.py · card hypotheses/H02-couplings-are-real/README.md · ratings: RE-A2 re-evaluation agent (suggested), round 1b

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b, Verdict corrected null)
SP1-SP5 synthetic recovery	AUC 0.95; leader top-1 1.00 at 4 of 5 $J_L \geq 0.3$; FPR 5.8–7.6%
RP1 mode C $\geq 1.5 \times$ mode I, above floor	6–8% → 5–11%; C/I 0.63, $\times$ 1.40
RP4 pairwise beats M1 on held-out days	2/21 → 3/21 chunks vs N1 $\times$
RP6 same-lab enrichment ≥ 1.3 ; RP7 #26 leader top 3	1.16 → 0.85; 8/10 → 10/10 $\times$
MF-P1 $\beta J_{0q} < 0.5$ every-where	max 0.36 → 0.34 $\checkmark$
MF-P2 III-C significant, mode I null	III-C 5/5 → 2/5; regime I 7/13 partial
MF-P3 forward $P(K)$ (unfitted)	CW better 5/13 → 6/9 partial
#45 holdout (rank 1, $z \geq 2$)	4/18, $z = 0.76$ (buggy table; $\times$ not re-run)
Native: #12 teams; 44b leader; NE14	$z = -0.32$; rank 1/5, $z = \times \times \times$ 0.94; rise → -0.08

GOAL PERIODS IN DETAIL

G45 ? 1b: not re-run (locked holdout; the executed run read... (**holdout**)
G10 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G12 $\times$ failed
G13 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G17 $\checkmark$ 1b: supported corrected data and DQ8 null: significant-c...
G18 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G19 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G20 $\checkmark$ 1b: supported corrected data and DQ8 null: significant-c...
G24 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G25 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
+8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The corrected null still leaks under start-up bursts (DQ8 size 0.51 for day-edge drives after trimming); H38's agent-state conditioning agrees with it here. Trimming keeps 33–99% of minutes (#44 only 33%). The #45 run used the buggy table; the thinning check draws day retention from non-holdout regime-III days, not #45's own. Native tests are small: 295 debate-minutes for #12, two days and five agents for 44b. The talk-spin result is post hoc. The synthetic harness was not re-run with the corrected calibration (active 0.67, J_{self} 0.49).

ROUND 2

Measure influence where it lives: H02-R1 (context-ledger visibility of messages), and pre-register the talk-spin coupling (regime-III shared-objective weeks) before any confirmatory use; whether to re-run the #45 rule on the fixed table is Vivian's call.

- **Essence**. Who changes whose mind?
- **H02-R1** Influence is the change in j 's next decision when i 's message is in j 's context (context ledger), compared with matched turns where it is not.
- **H02-R2** The #45 leader influenced plans, not activity: assignment-acceptance events (Jev-labeled) show its authority.
- **H02-R3** Influence follows authority markers (operator endorsement, titles, a leader role), not model family.

Second figure. (a) z of the collective coupling βJ_0 per chunk on the corrected grid: against the round-1 whole-day null almost every chunk looks coupled; against the corrected null (day edges trimmed, stalls masked) regime III collapses except #44. (b) Synthetic: the event-drop bug cut the #45 rule's power ($0.90 \rightarrow 0.70$ at $J_L = 0.3$) but would rarely turn a strong leader into $z = 0.76$.

